entire matter of the sun condensed into the space of an atom: a Planck star should be constituted by this extreme state of matter.

A Planck star is not stable: once compressed to the maximum it rebounds and begins to expand again. This leads to an explosion of the black hole. This process, as seen by a hypothetical observer sitting in the black hole on the Planck star, would be a rebound occurring at great speed. But time does not pass at the same speed for her as for those outside the black hole, for the same reason that in the mountains time passes faster than at sea-level. Except that for her, because of the extreme conditions, the difference in the passage of time is enormous, and what for the observer on the star would seem an extremely rapid bounce would appear, seen from outside it, to take place over a very long time. This is why we observe black holes remaining the same for long periods of time: a black hole is a rebounding star seen in extreme slow motion.

It is possible that in the furnace of the first instants of the universe black holes were formed, and that some of these are now exploding. If that were true, we could perhaps observe the signals which they emit when exploding, in the form of high-energy cosmic rays coming from the sky, thereby allowing us to observe and measure a direct effect of a phenomenon governed by quantum gravity. It's a bold idea – it might not work, for example, if in the primordial universe not enough black holes were formed to allow us to detect their explosions today. But the search for signals has begun. We shall see.

Another of the consequences of the theory, and one of the most spectacular, concerns the origins of the universe. We know how to reconstruct the history of our planet back to an initial period when it was tiny in size. But what about before that? Well, the equations of loop theory allow us to go even further back in the reconstruction of that history.

What we find is that when the universe is extremely compressed quantum theory generates a repulsive force, with the result that the great explosion or 'Big Bang' may have actually been a 'Big Bounce'. Our world may have actually been born from a preceding universe which contracted under its own weight until it was squeezed into a tiny space before 'bouncing' out and beginning to re-expand, thus becoming the expanding universe which we observe around us.

The moment of this bounce, when the universe was contracted into a nutshell, is the true realm of quantum gravity: time and space have disappeared altogether, and the world has dissolved into a swarming cloud of